

I like building things.

I'm a computer engineering student and undergraduate researcher at the University of Georgia. My work spans embedded hardware, robotics, and systems software, from a motor controller PCB and firmware to Windows driver and backend development.

CURRENT FOCUS

SERS lab research

Embedded systems

Systems programming

Robotics & autonomy

Selected Projects

Embedded hardware, backend systems, robotics, and computer vision.

EMBEDDED SYSTEMS · MOTOR CONTROL

01

Brushless Motor Controller

C · STM32 · KiCad · SPI

Designed a three-phase motor controller PCB in KiCad and wrote C firmware for PWM, SVPWM, and field-oriented control. The board has USB-C connectivity and supports an optional high-resolution SPI absolute encoder.

Verified PWM timing and deadtime with an oscilloscope, then debugged gate-drive and motor-control behavior on the assembled board.

Firmware PCB Design FOC SVPWM Hardware Debugging

[View project →](#)

SYSTEMS SOFTWARE · SECURITY

02

Soltix.cc

Founder · Four-year independent venture

Founded a commercial cheat-detection service that analyzed Windows process memory and system artifacts. It reached more than 300 users and 37 customers.

Built a Windows driver in C/C++ and process-analysis tools in Go, and administered Linux systems for customer deployments.

Windows Driver Process Memory C/C++ Go Linux

[View project →](#)

ROBOTICS · AUTONOMY

03

FRC Team 1683

Programming Lead · 2 Years · Team Member · 4 Years

Led programming for two of my four years on the team, developing autonomous and computer-vision software for competition robots. The 85-student team reached the FIRST World Championship in both years I served as lead.

Also contributed to robot design, electrical systems, fabrication, and CNC machining.

Robot Software Autonomous Systems Computer Vision
Electrical Systems CNC

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BACKEND DEVELOPMENT · HACKGT 13

04

SideQuests

MongoDB Prize · 1st Place, NSA HEARSAY

Built the Go backend and server infrastructure for a MongoDB-backed app that combines time- and location-aware activity plans with shared outings, interest matching, and group chat.

Go MongoDB Backend Development Server Infrastructure

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COMPUTER VISION · MECHANISM DESIGN

05

Sortify

Google Gemini Track Winner · MakeMIT x Harvard

Developed computer-vision software to classify waste and pass the results to sorting controls. Designed a compact differential mechanism to route each item to the correct bin.

Computer Vision Sorting Controls Mechanism Design

[View project](#) → [Devpost](#) ↗

BACKEND DEVELOPMENT · DOCUMENT SEARCH

06

DocDoctor

MongoDB Track Winner · Google Cloud Honorable Mention · AI ATL

Built a Go backend and WebSocket pipeline for live support-call transcripts and agent annotations. Added MongoDB persistence and document-search APIs so agents could retrieve relevant material during calls.

Go WebSockets MongoDB Document Search APIs

[View project](#) → [Devpost](#) ↗

AI · EDUCATION

07

Sophi

NexHacks · Carnegie Mellon

Built an AI learning tool that generates instructor-style questions from course material and gives targeted hints to help students practice problem solving with material tailored to their class.

AI Learning Tools Education

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ROBOTICS · COMPUTER VISION

08

Distributed Planetary Analysis Platform

3rd Place · Autonomous Track · RoboTech

Developed software for a humanoid robot and autonomous vehicle using inverse kinematics, fiducial detection, pathfinding, PWM motor/servo control, and obstacle avoidance. The autonomous vehicle ran on a Jetson and used fiducial tracking to navigate its environment.

Jetson Computer Vision Inverse Kinematics Pathfinding

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Experience

B.S. in Computer Engineering at the University of Georgia, expected May 2028. GPA: 3.9/4.0.

01	UGA SERS Lab	Undergraduate Researcher · Sep 2026–Present. Designed the scanner architecture and modeled the mechanical system in Onshape for an automated Raman/SERS platform. Selected electronics, corrected wiring faults, and contributed to fabrication and scanner software.
02	Soltix.cc	Founder · Four-year independent venture. Built commercial Windows process-analysis software, including a C/C++ driver and Go tools, and managed Linux deployments for 37 customers.
03	FRC Team 1683	Programming Lead · Two years. Led autonomous and computer-vision development on an 85-student team that reached the FIRST World Championship in both years I served as lead.

Technical Skills

Tools and disciplines used in my research, projects, and engineering work.

Languages

C, C++, Go, Python, Java, JavaScript

Engineering

Embedded systems, firmware, PCB design, electronics, robotics, CAD/CAM, Onshape

Software & systems

Windows driver development, computer vision, backend development, WebSockets, MongoDB, Linux

Maxim Iliev

University of Georgia · B.S. Computer Engineering · Expected May 2028

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